

WildfireVLM: AI-powered Analysis for Early Wildfire Detection and Risk Assessment Using Satellite Imagery

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[Code & Dataset](#)



Background



Escalating Wildfires

Wildfires are increasing in frequency and intensity due to climate change, causing significant loss of life, ecological harm, and massive CO₂ emissions.



System Limitations

Current detection systems are slow, resource-intensive, and prone to delays and inadequate risk assessment.



Motivation



Early Detection

Develop a system that identifies wildfires and smoke at their earliest stages to prevent escalation and minimize damage.



Risk Classification

Classify risks into **low, medium, or high** levels, enabling teams to allocate resources effectively.



Sustainability

Reduce dependency on expensive manual monitoring to make wildfire detection affordable and sustainable.



Related Work

Gonçalves et al.

Demonstrated the adaptability of **YOLO models** for detecting smoke and wildfire across diverse ground and aerial environments.

Kim & Muminov

Leveraged **YOLOv7 with UAV imagery** to capture high-resolution data over vast areas for early smoke detection.

Maillard et al.

Introduced **YOLO-NAS**, utilizing neural architecture search to balance wildfire detection accuracy with model efficiency.

Yang et al.

Enhanced smoke detection in UAV imagery by applying **transfer learning** to an improved YOLO v5 model.

Saydirasulovich

Developed an optimized **YOLO v8-based model** specifically designed to detect subtle wildfire smoke features in UAV footage.

Dataset Description

01

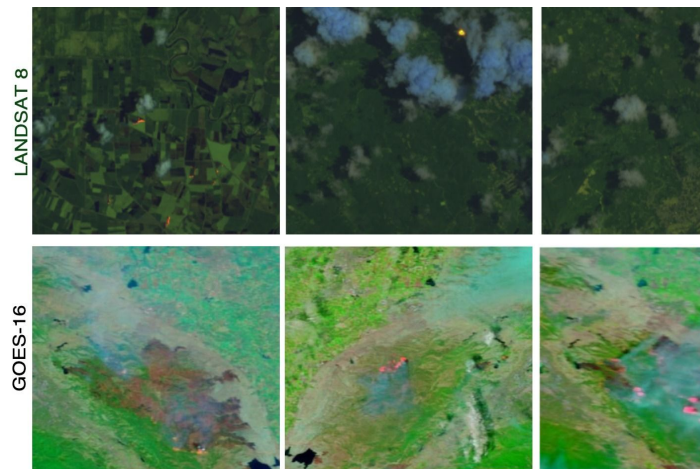
LANDSAT-8 : Captures wildfire imagery using its Operational Land Imager (OLI) and Thermal Infrared Sensor (TIRS), providing 15-30 meters spatial resolution for detecting fire-affected areas.

02

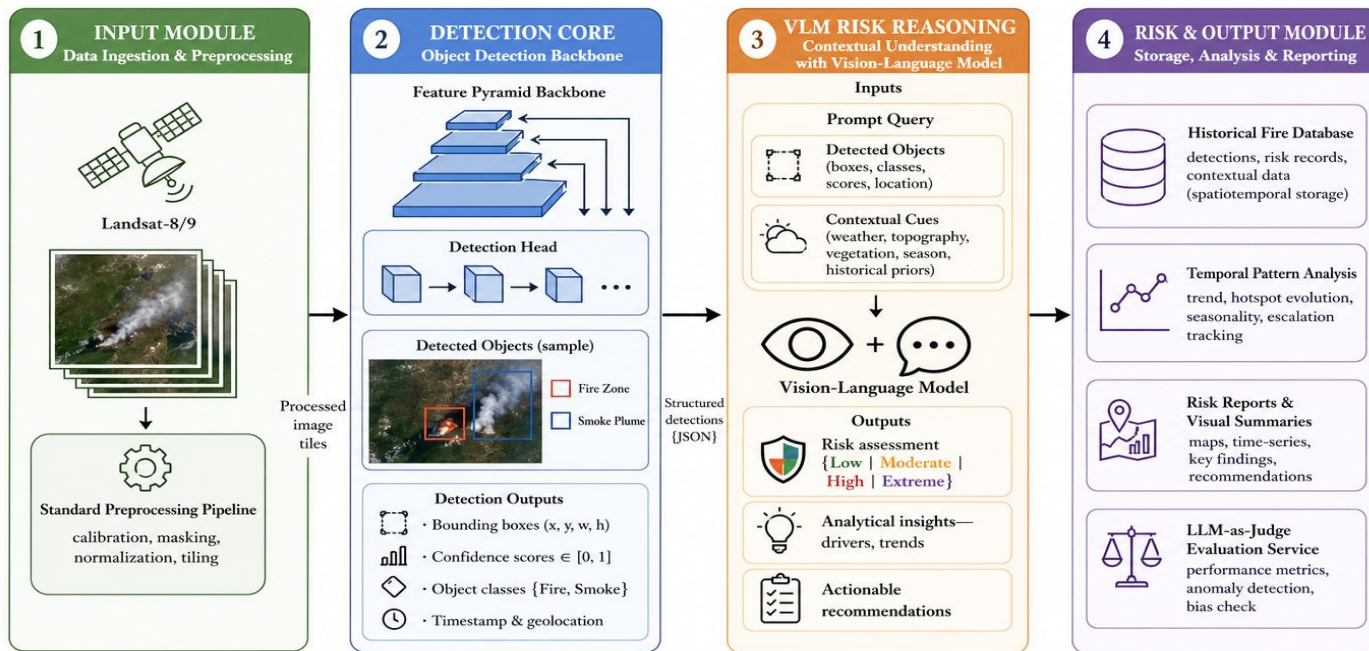
GOES-16 :The Geostationary Operational Environmental Satellites (GOES)-16 Launched in 2017, is a NASA and NOAA weather satellite providing high-frequency Earth imagery every 5 minutes across 16 spectral bands.

03

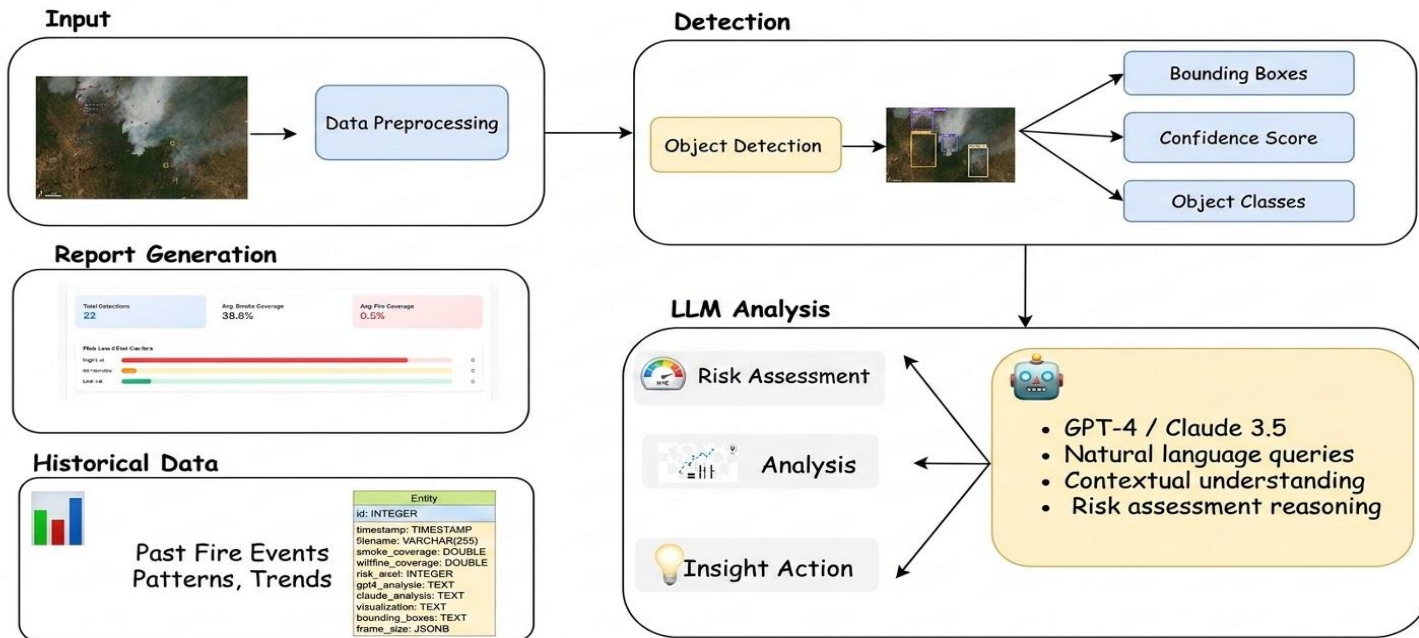
Preprocessing: Masking, calibrating and Annotating **3,771** Landsat-8 and GOES-16 satellite images into 2 classes: wildfire and smoke which is



Proposed Pipeline



End-to-End Workflow

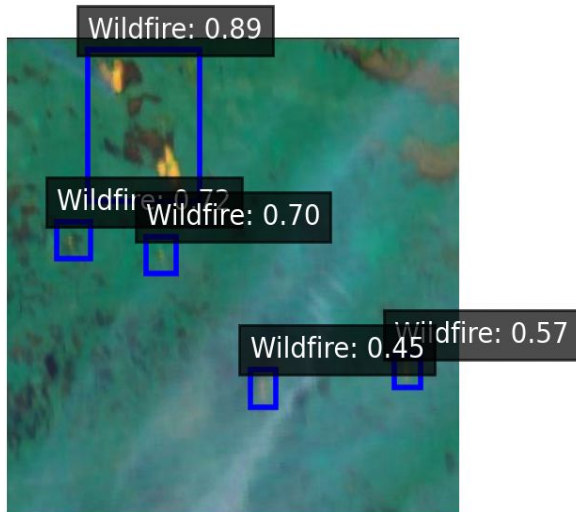


Quantitative Results

- **YOLO V11** : Robust performance in terms of localization
- **YOLO V12**: less false positive.

Model	mAP (%)	Precision (%)	Recall (%)	F1-score (%)
YOLO v8	72.1	60.7	67.6	64.0
YOLO 11	84.1	51.7	89.8	65.6
YOLO-NAS	54.1	56.0	57.1	56.6
YOLO v12	74.7	81.1	74.8	77.8

Risk Assessment



GPT-4o Assessment

Level 2 (Moderate) | 85% Conf.

Detections: 5 wildfires (conf. 0.45–0.89) • No smoke

Coverage: 9.81% • Scattered pattern

Growth: Moderate rate, dispersed ignition

Recommendations:

- **Deploy Resources:** Immediate firefighting action.
- **Surveillance:** Continuous monitoring.
- **Mobilize Units:** Ready aerial & ground units.

Claude 3.5 Assessment

Level 2 (Moderate) | 75% Conf.

Detections: 5 wildfires (conf. 0.40–0.90) • No smoke

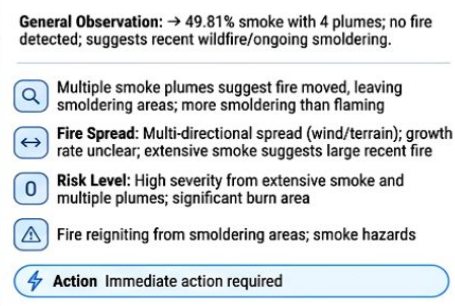
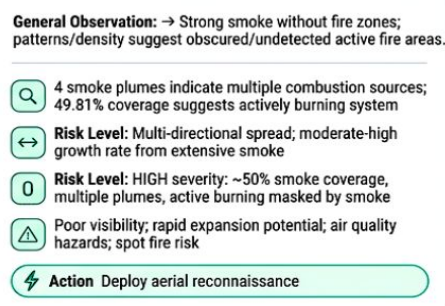
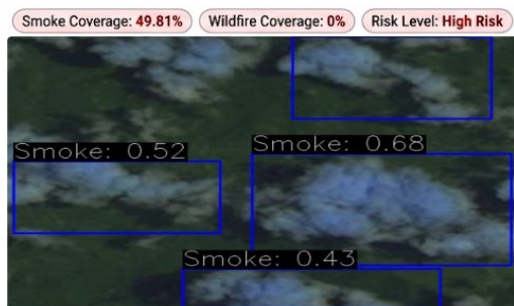
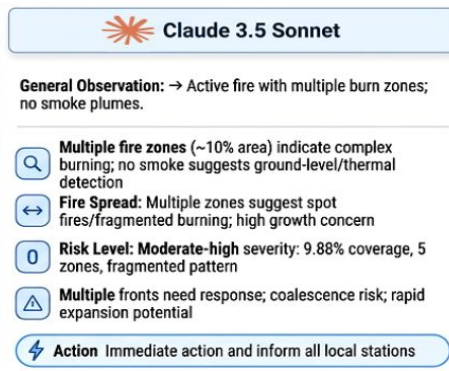
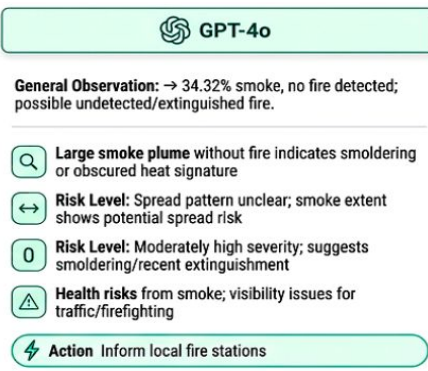
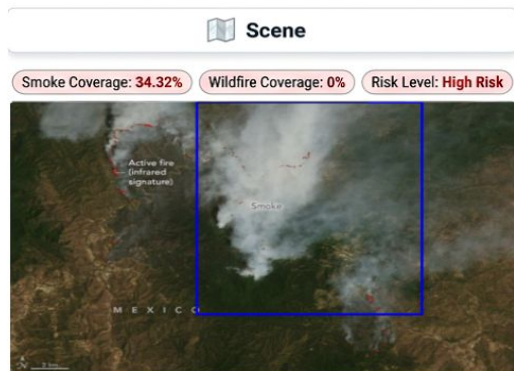
Coverage: 9.81% of total area (largest: 5.6%)

Growth: Scattered distribution, moderate potential

Recommendations:

- **Dispatch Crews:** Investigate and contain active locations.
- **Enhance Monitoring:** Increase aerial imaging frequency.
- **Resource Readiness:** Prepare firefighting assets.

Risk Assessment(Cont.)



LLM-as-a-Judge Framework

GPT-4o 

7.03 / 10

- Higher semantic accuracy
- Stronger actionable clarity
- More comprehensive risk assessment

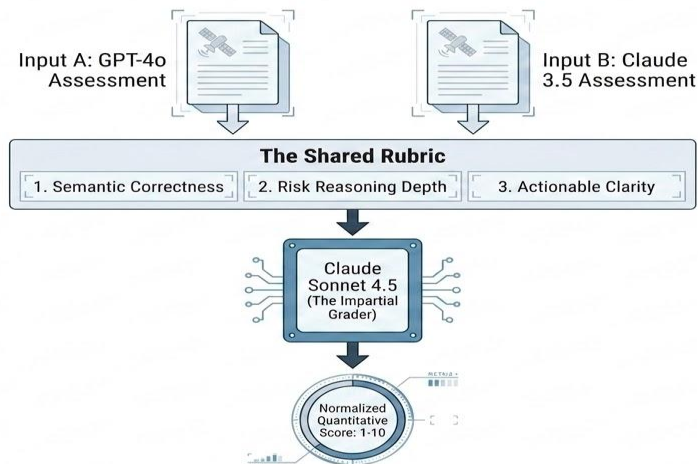
Claude 3.5 Sonnet 

6.16 / 10

- Reliable risk analysis
- Lower semantic accuracy & actionable clarity than GPT-4o

Evaluated both models using a **shared rubric** for semantic correctness, risk reasoning, and actionable clarity.

Overall: GPT-4o showed **stronger** alignment with the evaluation criteria.



Conclusion



End-to-end framework: Combines satellite imagery, **Object detection**, and VLM reasoning for wildfire analysis.



Object detection: YOLO v12 achieves **81.1% precision** and **77.8% F1-score** on 3,771 annotated images.



Beyond detection: VLMs transform detections into **contextual risk assessments and actionable recommendations**.



Model evaluation: GPT-4o achieves a higher mean judge score (**7.03** vs. **6.16**) than Claude 3.5 Sonnet.

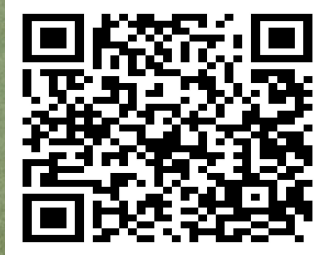
Future Work

- 01 Lightweight MLLM Integration:** Integrate lightweight Multimodal Language Models to significantly reduce response times.
- 02** Expand **expert-labeled assessments** across wildfire conditions and regions.
- 03 Efficient Fine-Tuning:** Fine-tune Vision-Language Models leveraging Low-Rank Adaptation (LoRA) and Q-LoRA frameworks.
- 04 Campfire Detection:** Identify and integrate unattended campfires, which account for more than 85% of all wildfires.
- 05 Advanced Risk Metrics:** Train on critical ecological factors (AQI, PM2.5, and PM10 indices) to enhance predictive risk calculations.

References

- 01 Zhao, Hongyang, et al. **"FSDF: A high-performance fire detection framework."** Expert Systems with Applications 238 (2024): 121665.
- 02 Cao, Lei, Zirui Shen, and Sheng Xu. **"Efficient forest fire detection based on an improved YOLO model."** Visual Intelligence 2.1 (2024): 20.
- 03 Rashkovetsky, Dmitry, et al. **"Wildfire detection from multisensor satellite imagery using deep semantic segmentation."** IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing 14 (2021): 7001-7016
- 04 Toan, N. T., Cong, P. T., Hung, N. Q. V., & Jo, J. (2019, November). **A deep learning approach for early wildfire detection from hyperspectral satellite images.** In 2019 7th International Conference on Robot Intelligence Technology and Applications (RiTA) (pp. 38-45). IEEE.

Thank you



Code & Dataset

